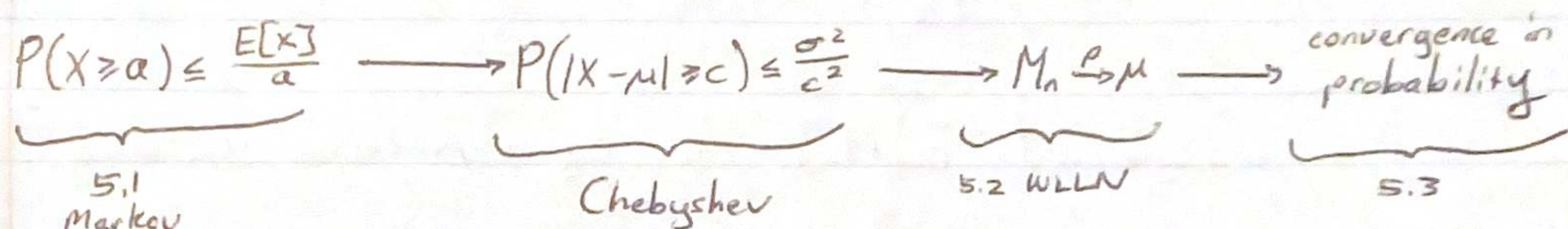


- RES.6-012 Introduction to Probability - Lectures 17, 18
Textbook Notes - Sections 5.1, 5.2, 5.3

Architecture



Section 5.1 Markov and Chebyshev Inequalities

Markov Inequality

For any non-negative RV X and $a > 0$:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Limitation: only uses mean; bounds are loose but require minimal information.

Chebyshev Inequality

For any RV X with mean μ and variance σ^2 , and for $c > 0$:

$$P(|X - \mu| \geq c) \leq \frac{\sigma^2}{c^2}$$

Equivalent form: (setting $c = k\sigma$):

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

↳ Probability of exceeding k standard deviations from mean is at most $1/k^2$.

Useful bound: if $X \in [a, b]$, then $\sigma^2 \leq (b-a)^2/4$, so even without knowing σ^2 :

$$P(|X - \mu| \geq c) \leq \frac{(b-a)^2}{4c^2}$$

• Section 5.2: Weak Law of Large Numbers

• Setup:

$X_1, X_2, \dots$ are i.i.d with mean μ and variance σ^2 .

Sample mean is:

$$M_n = \frac{X_1 + \dots + X_n}{n}$$

M_n is unbiased: $E[M_n] = \mu$. By independence: $\text{var}(M_n) = \frac{\sigma^2}{n} \rightarrow 0$

• WLLN

$$P(|M_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{n\epsilon^2} \rightarrow 0 \quad \text{as } n \rightarrow \infty, \text{ for any } \epsilon > 0$$

• Apply Chebyshev to M_n directly:

$$\text{var}(M_n) = \frac{\sigma^2}{n}, \text{ so } P(|M_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{(n\epsilon^2)}$$

• Interpretation $\rightarrow$ the sample mean concentrates around μ for large n .

Frequencies converge to probabilities.

Polling estimates converge to the true fraction.

Practical use (Polling Example)

With $\sigma^2 \leq \frac{1}{4}$ (Bernoulli worst case) and $P(|M_n - p| \geq \epsilon) \leq \frac{1}{(4n\epsilon^2)}$

$\rightarrow$ To guarantee error $\leq \epsilon$ with confidence $1 - \delta$: $n \geq \frac{1}{(4\delta\epsilon^2)}$

• Section 5.3 Convergence in Probability

• Definition

A sequence of RVs $Y_1, Y_2, \dots$ converges in probability to a constant a if:

$$\lim_{n \rightarrow \infty} P(|Y_n - a| \geq \epsilon) = 0 \quad \text{for every } \epsilon > 0$$

The WLLN is the statement that M_n converges in probability to μ .

More generally: if $E[Y_n] = \mu$ for all n and $\text{var}(Y_n) \rightarrow 0$, then $Y_n \xrightarrow{P} \mu$.

• Convergence in Probability vs. Convergence with Probability 1

Convergence with probability 1 (almost sure convergence):

$$P(\lim_{n \rightarrow \infty} Y_n = a) = 1$$

This is stronger: it states the sequence converges for essentially every outcome of the experiment, not just that deviations become unlikely.

Type	Statement	Used in
In probability	$P(\ Y_n - a\ \geq \epsilon) \rightarrow 0$	WLLN
With probability 1	$P(\lim Y_n = a) = 1$	SLLN (section 5.5)

• Convergence w.p.1 $\Rightarrow$ convergence in probability. (Converse fails)

• Counter Example (E.g. 5.8): $Y_n = n^2$ with probability $\frac{1}{n}$, and 0 otherwise.

• $Y_n \rightarrow 0$ in probability, but $E[Y_n] = n \rightarrow \infty$ (expectations need not converge even when RVs do.)